EXPERIMENT 09

Rule-Based Part-of-Speech Tagging using Regular Expressions

AIM

To construct a rule-based Part-of-Speech tagger using regular expression patterns to classify tokens into grammatical categories in Python.

PROCEDURE

1. Identify morphological regularities (e.g. affixes, capitalization, numeric patterns) associated with POS categories.
2. Formulate sequential regular expression rules for gerunds (-ing), past tense (-ed), adverbs (-ly), plurals (-s), numerals, and determiners.
3. Utilize NLTK's `RegexpTagger` to compile the ordered pattern hierarchy.
4. Tag test sentences and evaluate rule matches.
5. Display the labeled tokens and the corresponding linguistic rules applied.

PROGRAM

```
import nltk
from nltk.tag import RegexpTagger

patterns = [
    (r'.*ing$', 'VBG'),
    (r'.*ed$', 'VBD'),
    (r'.*es$', 'VBZ'),
    (r'.*ould$', 'MD'),
    (r'.*'s$', 'NN$'),
    (r'.*s$', 'NNS'),
    (r'^-?[0-9]+(\.[0-9]+)?$', 'CD'),
    (r'^(the|a|an|this|that)$', 'DT'),
    (r'.*ly$', 'RB'),
    (r'.*able$', 'JJ'),
    (r'^[A-Z].*', 'NNP'),
    (r'.*', 'NN')
]

regex_tagger = RegexpTagger(patterns)

test_sentence = "The quickly running programmer completed 42 complex tasks in
Paris .".split()
tagged = regex_tagger.tag(test_sentence)

print(f"{'Token':<14} | {'Assigned Tag':<10} | {'Morphological Pattern Rule'}")
print("-" * 55)
for word, tag in tagged:
    rule_desc = {
```

```
        'DT': 'Exact Determiner match',
        'RB': 'Suffix -ly (Adverb)',
        'VBG': 'Suffix -ing (Gerund)',
        'VBD': 'Suffix -ed (Past Verb)',
        'CD': 'Numeric Digit pattern',
        'NNS': 'Suffix -s (Plural)',
        'NNP': 'Capitalized (Proper Noun)',
        'JJ': 'Adjective Pattern',
        'NN': 'Default Fallback (Noun)'
    }.get(tag, 'Rule Match')
    print(f"{word:<14} | {tag:<10} | {rule_desc}")
```

OUTPUT

```
> python3 experiment_09.py
-----------------------------------------------------------------------------------------------------------
Token          | Assigned Tag | Morphological Pattern Rule
-----------------------------------------------------------
The            | DT         | Exact Determiner match
quickly        | RB         | Suffix -ly (Adverb)
running        | VBG        | Suffix -ing (Gerund)
programmer     | NN         | Default Fallback (Noun)
completed      | VBD        | Suffix -ed (Past Verb)
42             | CD         | Numeric Digit pattern
complex        | NN         | Default Fallback (Noun)
tasks          | NNS        | Suffix -s (Plural)
in             | NN         | Default Fallback (Noun)
Paris          | NNP        | Capitalized (Proper Noun)
.              | NN         | Default Fallback (Noun)
[Process completed with exit code 0]
```

RESULT

Thus, the rule-based Part-of-Speech tagging system using regular expression patterns was successfully implemented and evaluated.